AS1109 - Solved Practice Set I

1. Limits and continuity

(a) Evaluate

$$\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}}.$$

(b) For $(x,y) \neq (0,0)$ let

$$g(x,y) = \frac{\sqrt{9+x^2+y^2}-3}{x^2+y^2}.$$

Choose $g(0,0)$ so that g is continuous at the origin.

Solution

(a) Since

$$2|xy| \le x^2+y^2,$$

$$\left|\frac{xy}{\sqrt{x^2+y^2}}\right| \le \frac{\sqrt{x^2+y^2}}{2} \to 0.$$

Hence

$$\boxed{0}.$$

(b) Rationalising,

$$g(x,y) = \frac{1}{\sqrt{9+x^2+y^2}+3}.$$

Therefore

$$\lim_{(x,y)\to(0,0)} g(x,y) = \frac{1}{6},$$

so

$$\boxed{g(0,0) = \frac{1}{6}}.$$

2. Functions, regions, ranges, inverses, compositions, traces and level curves

(a) For

$$F(x,y) = \frac{\sqrt{(y+1)(3-y)}}{1+x^2},$$

find the domain and range, and classify the domain.

(b) For

$$z = x^2 + 2y,$$

find the traces in $y=0$ and $x=0$, and describe the level curves $z=c$.

(c) Let $p(t) = e^t$ and $q(t) = 1+t^2$. Find p^{-1}, $(p\circ q)(t)$ and $(q\circ p)(t)$.

Solution

(a) The square root requires

$$(y+1)(3-y) \ge 0,$$

hence

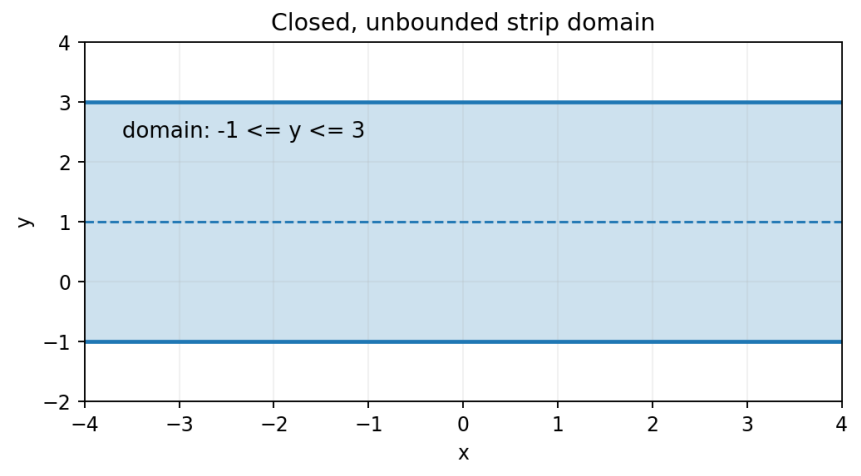
$$\boxed{-1 \le y \le 3}.$$

There is no restriction on x, so the domain is **closed and unbounded**.
Also,

$$(y+1)(3-y) = 4-(y-1)^2,$$

so the numerator lies in $[0,2]$, while $1+x^2 \ge 1$. Therefore

$$\boxed{\text{Range}(F) = [0,2]}.$$



(b) If $y = 0$, then

$$\boxed{z = x^2},$$

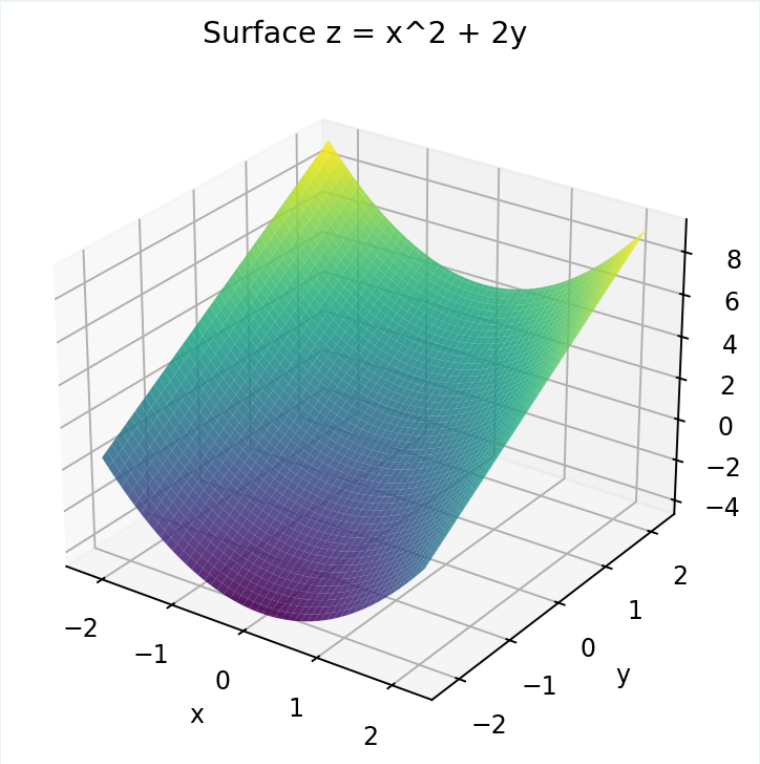
while if $x = 0$,

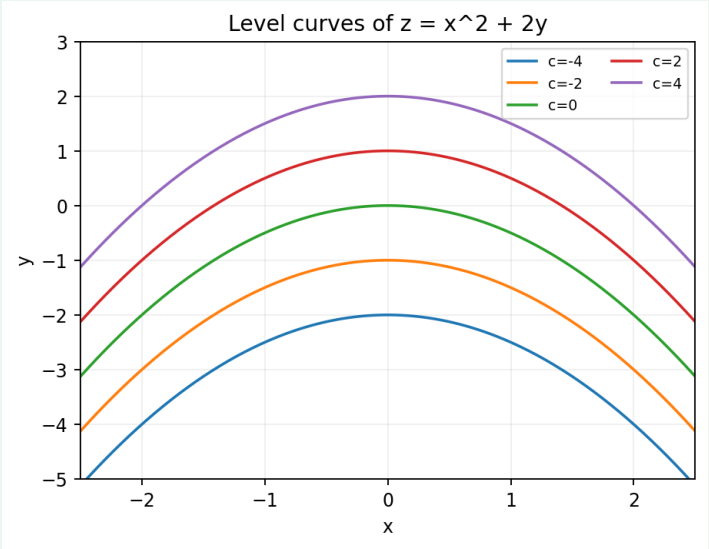
$$\boxed{z = 2y}.$$

For $z = c$,

$$x^2 + 2y = c \quad \Longrightarrow \quad \boxed{y = \frac{c - x^2}{2}},$$

so the level curves are downward-opening parabolas.





(c)

$$\boxed{p^{-1}(s) = \ln s,\ \ s > 0},$$

$$\boxed{(p \circ q)(t) = e^{1+t^2}}, \qquad \boxed{(q \circ p)(t) = 1 + e^{2t}}.$$

3. Partial differentiation and implicit partial differentiation

(a) For

$$f(x, y) = \ln(1 + x^2 + y^2) + x \sin y,$$

find f_x, f_y, f_{xy}, and evaluate $f_{xy}(1, 0)$.

(b) The equation

$$xe^z + yz = 2$$

defines $z = z(x, y)$ near $(0, 1, 2)$. Find z_x and z_y there.

Solution

(a)

$$f_x = \frac{2x}{1 + x^2 + y^2} + \sin y, \qquad f_y = \frac{2y}{1 + x^2 + y^2} + x \cos y.$$

Hence

$$f_{xy} = -\frac{4xy}{(1 + x^2 + y^2)^2} + \cos y,$$

so

$$\boxed{f_{xy}(1, 0) = 1}.$$

(b) Let $H(x, y, z) = xe^z + yz - 2$. Then

$$z_x = -\frac{H_x}{H_z}, \qquad z_y = -\frac{H_y}{H_z}.$$

Now

$$H_x = e^z, \qquad H_y = z, \qquad H_z = xe^z + y.$$

At $(0, 1, 2)$,

$$\boxed{z_x = -e^2, \qquad z_y = -2}.$$

4. Linearization

Find the linearization of

$$f(x, y) = (x^2 + y)^{1/3}$$

at $(2, 4)$, and use it to approximate $f(2.03, 3.96)$.

Solution

$$f(2,4) = 2,$$

$$f_x = \frac{2x}{3(x^2+y)^{2/3}}, \qquad f_y = \frac{1}{3(x^2+y)^{2/3}}.$$

Thus

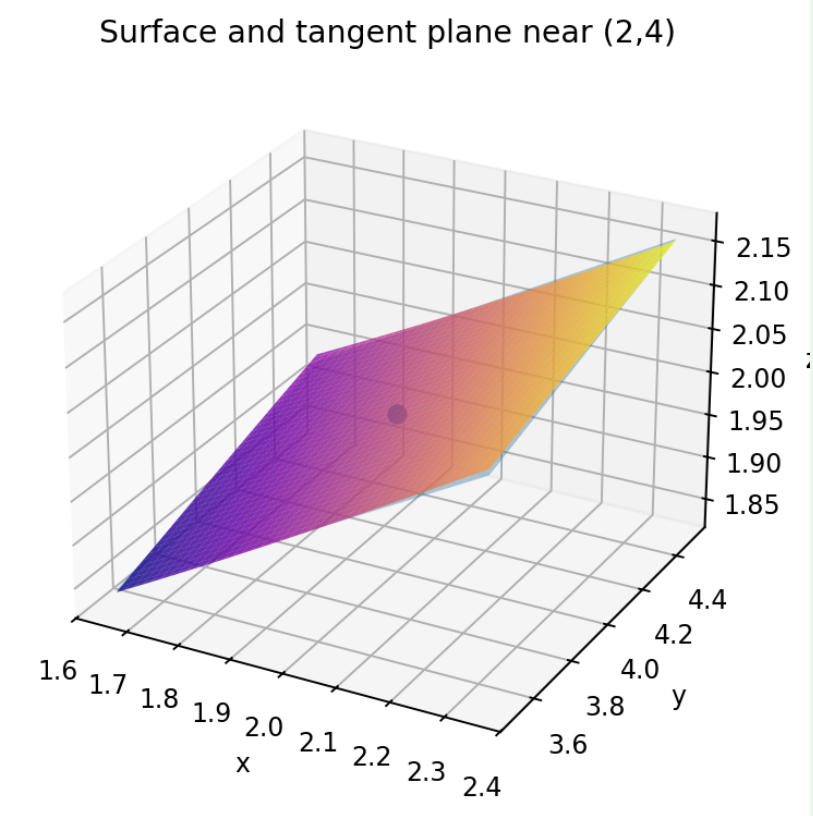
$$f_x(2,4) = \frac{1}{3}, \qquad f_y(2,4) = \frac{1}{12}.$$

Therefore

$$\boxed{L(x,y) = 2 + \frac{1}{3}(x-2) + \frac{1}{12}(y-4)}.$$

Using $\Delta x = 0.03$, $\Delta y = -0.04$,

$$f(2.03, 3.96) \approx 2 + \frac{1}{3}(0.03) + \frac{1}{12}(-0.04) = \boxed{2.0067}.$$



5. Sensitivity of fluid flow to pipe dimensions

Suppose

$$Q = K\frac{r^4 \Delta P}{L},$$

where K is constant. Derive a relative-change formula. Estimate the percentage change in Q if r increases by 0.8%, ΔP increases by 1.5%, and L increases by 0.4%.

Solution

Taking relative differentials,

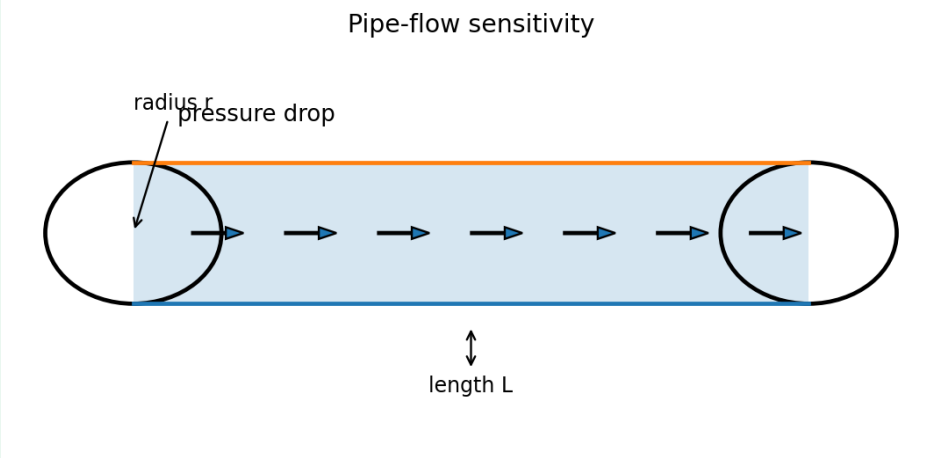
$$\boxed{\frac{dQ}{Q} = 4\frac{dr}{r} + \frac{d(\Delta P)}{\Delta P} - \frac{dL}{L}}.$$

Hence

$$\frac{dQ}{Q} \approx 4(0.8\%) + 1.5\% - 0.4\% = 4.3\%.$$

Therefore

$$\boxed{Q \text{ increases by approximately } 4.3\%.}$$



6. Multivariable chain rule

Let

$$z = \ln(1 + x^2 + y^2), \qquad x = u\cos v, \qquad y = u\sin v.$$

Find z_u and z_v using the chain rule.

Solution

$$z_x = \frac{2x}{1 + x^2 + y^2}, \qquad z_y = \frac{2y}{1 + x^2 + y^2}.$$

Also,

$$x_u = \cos v,\ y_u = \sin v, \qquad x_v = -u\sin v,\ y_v = u\cos v.$$

Therefore

$$z_u = z_x x_u + z_y y_u = \frac{2u}{1 + u^2},$$

and

$$z_v = z_x x_v + z_y y_v = 0.$$

Hence

$$\boxed{z_u = \frac{2u}{1 + u^2}, \qquad z_v = 0}.$$

Check

Direct substitution gives $z = \ln(1 + u^2)$.

7. Local and absolute extrema

(a) Find and classify all critical points of

$$f(x, y) = xe^{-(x^2+y^2)}.$$

(b) Find the absolute extrema of

$$g(x, y) = xy$$

on

$$x \geq 0, \qquad y \geq 0, \qquad x + y \leq 4.$$

Solution

(a)

$$f_x = e^{-(x^2+y^2)}(1 - 2x^2), \qquad f_y = -2xye^{-(x^2+y^2)}.$$

Thus

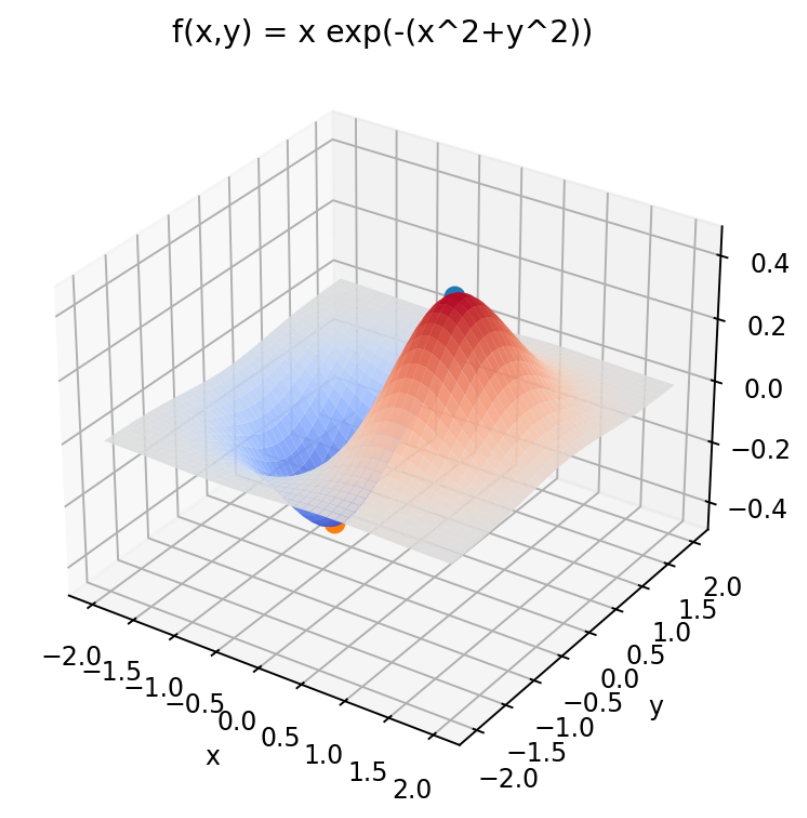
$$\left(\frac{1}{\sqrt{2}}, 0\right), \qquad \left(-\frac{1}{\sqrt{2}}, 0\right)$$

are the critical points. At both,

$$D = \frac{4}{e} > 0.$$

At $x = 1/\sqrt{2}$, $f_{xx} < 0$, so there is a local maximum; at $x = -1/\sqrt{2}$, $f_{xx} > 0$, so there is a local minimum:

$$\boxed{f_{\max} = \frac{1}{\sqrt{2e}}, \qquad f_{\min} = -\frac{1}{\sqrt{2e}}}.$$



(b) On $x = 0$ or $y = 0$, $g = 0$. On $x + y = 4$,

$$g = x(4 - x).$$

Thus

$$g' = 4 - 2x = 0 \quad \Longrightarrow \quad x = 2,\ y = 2,$$

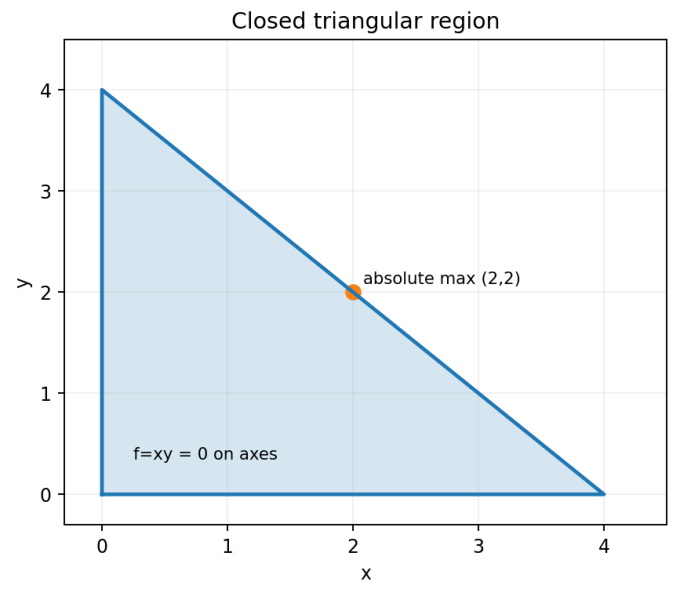
and

$$\boxed{g_{\max} = 4 \text{ at } (2, 2)}.$$

The minimum is

$$\boxed{g_{\min} = 0}$$

on the two coordinate-axis edges.



8. Arc length and an astroid segment

(a) Find the arc length of

$$y = \frac{x^3}{6} + \frac{1}{2x}, \qquad 1 \le x \le 2.$$

(b) For

$$x^{2/3} + y^{2/3} = 9,$$

find the first-quadrant arc length from $(8, 5\sqrt{5})$ to $(27, 0)$.

Solution

(a)

$$y' = \frac{x^2}{2} - \frac{1}{2x^2}.$$

Then

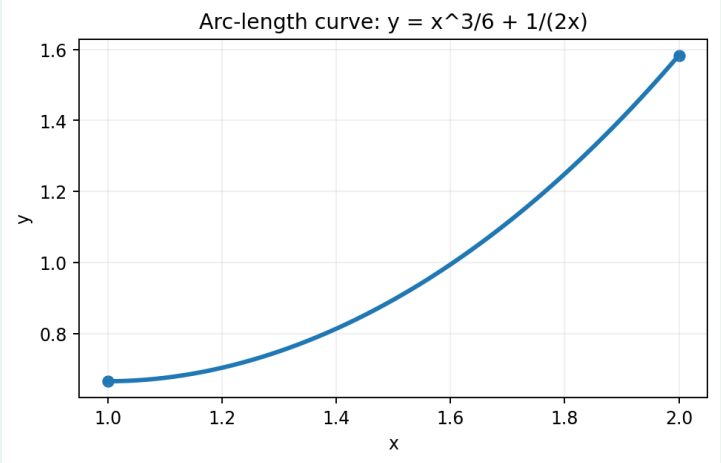
$$1 + (y')^2 = \frac{(x^4+1)^2}{4x^4},$$

so

$$\sqrt{1 + (y')^2} = \frac{1}{2}(x^2 + x^{-2}).$$

Therefore

$$L = \frac{1}{2}\int_1^2 (x^2 + x^{-2})\,dx = \left[\frac{x^3}{6} - \frac{1}{2x}\right]_1^2 = \boxed{\frac{17}{12}}.$$



(b) Implicit differentiation gives

$$y' = -\frac{y^{1/3}}{x^{1/3}}.$$

Hence

$$\sqrt{1+(y')^2} = 3x^{-1/3}.$$

Thus

$$L = \int_8^{27} 3x^{-1/3}\,dx = \left[\frac{9}{2}x^{2/3}\right]_8^{27} = \frac{9}{2}(9-4).$$

Therefore

$$\boxed{L = \frac{45}{2}}.$$

